Minecraft Operator Runbook

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0. Assumptions & Prerequisites

- AWS account with EC2, ECR, and S3 access
- AWS CLI configured
- Ubuntu EC2 instance
- SSH access available

1. Runtime Architecture

Internet → Security Group (25565) → EC2 → Docker → Minecraft Container → EBS Storage

- Port 25565 exposed
- Docker containerized runtime
- Persistent storage via bind mount

2. Initial Deployment

Install Dependencies

```
sudo apt update
sudo apt install -y docker.io docker-compose
sudo systemctl enable docker
sudo systemctl start docker
```

Authenticate to ECR

```
aws ecr get-login-password --region us-east-1 \
| docker login --username AWS --password-stdin <ECR_URL>
```

Start Service

```
docker compose up -d
```

3. Image Policy

- No `latest` tag
- Use: mc-<version>-build<num>
- Stored in private ECR

4. Upgrade & Rollback

Upgrade

```
docker compose up -d
```

Validation

```
docker compose ps
docker compose logs --tail=50
nmap -sV -Pn -p T:25565 localhost
```

Rollback

```
docker compose up -d
```

5. Backup & Restore

Backup

```
tar -czvf world-backup-$(date +%F).tar.gz data/
aws s3 cp world-backup.tar.gz s3://<BUCKET>/
```

Restore

```
docker compose down
rm -rf data/*
aws s3 cp backup.tar.gz .
tar -xzvf backup.tar.gz
docker compose up -d
```

6. Monitoring

```
docker compose ps
docker compose logs
docker stats
htop
```

7. Failure Scenarios

- Crash loop → check logs, rollback
- Port closed → check security group
- Connection issues → verify service + MOTD
- Corrupt data → restore backup

8. Security

- Restrict SSH access
- Use IAM roles
- Private S3 bucket

9. Video

https://media.oregonstate.edu/media/t/1_tloxm5xl

10. Attribution

itzg/minecraft-server, course repo, LLM assistance